

Noether's Theorem

Every continuous symmetry of the action is a conservation law. It is the most important theorem in this book, and the proof is half a page.

WHERE WE ARE

Conservation laws keep turning up, and every time they have been earned by a different trick. In Chapter 1.1 energy appeared because someone thought to dot $F = ma$ with $\mathbf{v}$; momentum appeared because Newton's third law made the internal forces cancel in pairs. In Chapter 0.8 the oscillator's energy appeared as a "first integral" found by inspection, and a three-mass chain turned out to have a zero-frequency mode that was really momentum conservation in disguise. In Chapter 1.2 the Beltrami identity produced a conserved E out of the assumption $\partial L / \partial t = 0$, and the sphere's geodesics produced Clairaut's relation out of the absence of ϕ from the integrand. Five separate conjuring tricks.

They are one theorem. Emmy Noether proved it in 1918, and its content is not a computational convenience. It is this: **conservation laws are not facts about forces. They are facts about symmetry.** Momentum is conserved because space is the same here as over there. Energy is conserved because time is the same now as it was yesterday. Neither statement mentions what the matter is made of, and neither can be broken by discovering a new force — only by discovering that space or time is not uniform.

That is worth the price of admission on its own. But it is not why this chapter is the hinge of the book. Read the theorem backwards and it stops being a way of *finding* conservation laws and becomes a way of *building theories*: if you know in advance what is conserved, you know what symmetry the action must have, and demanding that symmetry is often enough to determine the action almost uniquely. From Part II onward that is the entire method. We will not discover the Dirac equation or the Standard Model by looking at data and guessing. We will write down a symmetry group, ask for the most general action invariant under it, and read off the physics. Chapter 1.2 gave us the machine that turns an action into equations of motion. This chapter gives us the only known reason to prefer one action over another.

It also closes Part I. Four chapters ago mechanics was a list of forces.

TOOLS YOU'LL NEED — Chapter 0.6: the multivariable chain rule, and the equality of mixed partial derivatives (the proof turns on it twice). Chapter 0.7: the continuity equation $\partial_t \rho + \nabla \cdot \mathbf{J} = 0$ and the divergence theorem — §6 shows that the field version of this theorem produces exactly that equation. Chapter 1.2: the action $S = \int L dt$, the Euler–Lagrange equation, and above all its Problem 4 — adding a total time derivative $\frac{dF}{dt}$ to L changes nothing physical. That freedom is not a footnote here; it is part of the hypothesis of the theorem, and §1 shows a symmetry that fails without it. Chapter 1.3: the Poisson bracket and the fact that a phase-space function generates an infinitesimal transformation. §7 needs it, and states what it needs.

1 · What a symmetry is, precisely

The word "symmetry" in ordinary speech means a shape looks the same after you flip it. The word in physics means something narrower and more useful, and getting the definition exactly right is most of the work — the proof in §2 is four lines once the hypothesis is stated properly.

1.1 · A continuous family of transformations

Start with a system described by coordinates $q_1(t), \dots, q_n(t)$. A **continuous transformation** is a recipe that takes any path and deforms it, smoothly, by an amount controlled by a single real parameter ϵ :

$$q_i(t) \mapsto q_i(t, \epsilon), \quad \text{with} \quad q_i(t, 0) = q_i(t). \quad (1.4.1)$$

The condition $q_i(t, 0) = q_i(t)$ says that $\epsilon = 0$ is the do-nothing transformation. That is what makes the family *continuous* in the sense we need: we can turn the transformation on gradually, and therefore we can differentiate with respect to how much of it we have turned on.

Only the derivative at $\epsilon = 0$ will matter. Define

$$K_i(q, \dot{q}, t) \equiv \left. \frac{\partial q_i(t, \epsilon)}{\partial \epsilon} \right|_{\epsilon=0},$$

so that $\delta q_i \equiv q_i(t, \epsilon) - q_i(t) = \epsilon K_i + O(\epsilon^2)$. (1.4.2)

K_i is called the **generator** of the transformation — it is a vector field on configuration space telling you which way each coordinate moves when you nudge ϵ . Three examples, all of which we will use:

TRANSFORMATION	ACTING ON	GENERATOR K
Translate by ϵ along $\mathbf{n}$	$\mathbf{x} \mapsto \mathbf{x} + \epsilon \mathbf{n}$	$\mathbf{K} = \mathbf{n}$
Rotate by ϵ about $\mathbf{n}$	$\mathbf{x} \mapsto \mathbf{x} + \epsilon \mathbf{n} \times \mathbf{x} + O(\epsilon^2)$	$\mathbf{K} = \mathbf{n} \times \mathbf{x}$
Boost by $\epsilon \mathbf{u}$	$\mathbf{x} \mapsto \mathbf{x} + \epsilon \mathbf{u} t$	$\mathbf{K} = \mathbf{u} t$

The rotation entry deserves a second look, since it is the only one that is not obvious. A rotation by a small angle ϵ about the axis $\mathbf{n}$ moves the tip of $\mathbf{x}$ along a circle of radius $|\mathbf{x}| \sin \theta$, perpendicular to both $\mathbf{n}$ and $\mathbf{x}$, through an arc length $\epsilon |\mathbf{x}| \sin \theta$. The vector with exactly that direction and that magnitude is $\epsilon \mathbf{n} \times \mathbf{x}$. So the *infinitesimal* version of a rotation is a cross product, which is why angular momentum will turn out to be one.

1.2 · Invariance of the action — with one crucial loosening

Now: what should it mean for the transformation to be a symmetry? The naive answer is " L doesn't change". That answer is too strong, and if you adopt it you will conclude — wrongly — that Newtonian mechanics is not invariant under a change of reference velocity.

The right condition comes from remembering what L is *for*. Nothing physical depends on L directly; the physics is in the stationary points of $S = \int L dt$. And Chapter 1.2's Problem 4 established that two Lagrangians differing by a total time derivative,

$$L' = L + \frac{dF(q, t)}{dt}, \quad (1.4.3)$$

have *identical* Euler–Lagrange equations — because the extra term integrates to $F(q(t_2), t_2) - F(q(t_1), t_1)$, which is the same number for every path in the competition and therefore invisible to the variation. So a transformation that changes L by a total derivative has not changed the physics at all, and refusing to call it a symmetry would be a mistake of bookkeeping.

DEFINITION — A SYMMETRY OF THE ACTION

The family (1.4.1) is a **symmetry** of the Lagrangian L if there exists a function $F(q, t)$ such that

$$\delta L \equiv L(q + \delta q, \dot{q} + \delta \dot{q}, t) - L(q, \dot{q}, t) = \epsilon \frac{dF}{dt} + O(\epsilon^2)$$

for *every* path $q(t)$ — not merely for solutions of the equations of motion. The case $F = 0$ (" L is strictly invariant") is allowed and common; the general case is called *invariance up to a total derivative*, or quasi-invariance.

Two features of this definition will do real work later. First, it is a condition on the **action**, not on the equations of motion; §7 exhibits a transformation that preserves the equations of motion but not the action, and produces no conserved quantity. Second, the hypothesis is **off shell** — required for all paths — while the conclusion of §2 will hold only **on shell**, along actual trajectories. Getting those two quantifiers the wrong way round is the single most common way to misstate this theorem.

1.3 · Why the total derivative is not a technicality: Galilean boosts

Here is the case that forces the loosening. Take N particles interacting through a potential that depends only on their *separations* — which is every non-relativistic system of mutually interacting bodies you have ever seen:

$$L = \sum_{a=1}^N \frac{1}{2} m_a \dot{\mathbf{x}}_a \cdot \dot{\mathbf{x}}_a - V(\{\mathbf{x}_a - \mathbf{x}_b\}). \quad (1.4.4)$$

Now change to a reference frame moving at a small constant velocity $-\epsilon \mathbf{u}$. In the new frame every particle's position is shifted by $\epsilon \mathbf{u} t$:

$$\mathbf{x}_a \longmapsto \mathbf{x}_a + \epsilon \mathbf{u} t, \quad \dot{\mathbf{x}}_a \longmapsto \dot{\mathbf{x}}_a + \epsilon \mathbf{u}. \quad (1.4.5)$$

Galilean relativity — the statement that you cannot detect uniform motion by any mechanical experiment inside a sealed cabin — says this had better be a symmetry. Is it? The potential is untouched, since every separation $\mathbf{x}_a - \mathbf{x}_b$ is

unchanged by shifting *all* positions equally. But the kinetic energy is not:

$$\begin{aligned}\frac{1}{2}m_a |\dot{\mathbf{x}}_a + \epsilon \mathbf{u}|^2 &= \frac{1}{2}m_a |\dot{\mathbf{x}}_a|^2 + \epsilon m_a \dot{\mathbf{x}}_a \cdot \mathbf{u} + O(\epsilon^2), \\ \delta L &= \epsilon \mathbf{u} \cdot \sum_a m_a \dot{\mathbf{x}}_a + O(\epsilon^2).\end{aligned}\tag{1.4.6}$$

So $\delta L \neq 0$: the Lagrangian is *not* invariant under a boost. If "symmetry" meant " L unchanged" we would be forced to say that Newtonian mechanics violates Galilean relativity, which is absurd.

Look again at what δL actually is. The masses are constants, so

$$\mathbf{u} \cdot \sum_a m_a \dot{\mathbf{x}}_a = \frac{d}{dt} \left(\mathbf{u} \cdot \sum_a m_a \mathbf{x}_a \right) = \frac{dF}{dt}, \quad F = M \mathbf{u} \cdot \mathbf{R}_{\text{cm}},\tag{1.4.7}$$

where $M = \sum_a m_a$ and $\mathbf{R}_{\text{cm}} = \frac{1}{M} \sum_a m_a \mathbf{x}_a$ is the centre of mass. The change in L is a total time derivative — exactly the case the definition was widened to admit. A boost is a symmetry, in the only sense that matters, and it is not a symmetry in the naive sense.

The moral is worth stating flatly, because it will save you confusion in three later chapters: **if you ever find that a symmetry you believe in appears to fail, check whether δL is a total derivative before concluding anything.** Worked Example 1 extracts the conserved quantity that this particular F produces, and it is a conservation law that nobody thinks to name.

2 · The theorem, proved

Everything is now in place. The proof is short, and each step uses exactly one hypothesis, so it is worth watching which.

NOETHER'S THEOREM (TRANSFORMATIONS THAT DO NOT MOVE THE CLOCK)

Let $\delta q_i = \epsilon K_i$ be a symmetry of L in the sense of §1.2, with $\delta L = \epsilon \frac{dF}{dt}$. Then along every solution of the Euler–Lagrange equations the quantity

$$Q = \sum_i \frac{\partial L}{\partial \dot{q}_i} K_i - F$$

is constant in time.

2.1 · Step 1 — the two derivatives commute

We will need to know how the *velocities* respond, and there is a small fact to establish first. The transformed velocity is $\dot{q}_i(t, \epsilon) = \partial_t q_i(t, \epsilon)$, so

$$\delta \dot{q}_i = \epsilon \left. \frac{\partial}{\partial \epsilon} \frac{\partial q_i}{\partial t} \right|_{\epsilon=0} = \epsilon \left. \frac{\partial}{\partial t} \frac{\partial q_i}{\partial \epsilon} \right|_{\epsilon=0} = \epsilon \frac{dK_i}{dt} = \frac{d}{dt}(\delta q_i). \quad (1.4.8)$$

The middle equality is the equality of mixed partial derivatives (Chapter 0.6). In words: **varying and then differentiating in time gives the same thing as differentiating and then varying**. That is only true because δ is taken at *fixed* t — and it is precisely why §2.4, where the transformation moves t as well, needs extra care rather than more of the same.

2.2 · Step 2 — expand δL by the chain rule

L is a function of $2n + 1$ slots. Nudging the first $2n$ of them and applying the multivariable chain rule of Chapter 0.6:

$$\delta L = \sum_i \left[\frac{\partial L}{\partial q_i} \delta q_i + \frac{\partial L}{\partial \dot{q}_i} \delta \dot{q}_i \right] = \epsilon \sum_i \left[\frac{\partial L}{\partial q_i} K_i + \frac{\partial L}{\partial \dot{q}_i} \frac{dK_i}{dt} \right], \quad (1.4.9)$$

using Step 1 in the second equality. Note that t itself has not moved, so there is no $\partial L / \partial t$ term. Nothing has been assumed yet except smoothness; equation (1.4.9) is true for every path.

2.3 · Step 3 — use the equations of motion

Now, and only now, restrict attention to a path that actually solves the Euler–Lagrange equation,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i}. \quad (1.4.10)$$

Substitute the left side of (1.4.10) in place of $\partial L / \partial q_i$ in (1.4.9). The bracket becomes something you have seen before:

$$\begin{aligned} \delta L &= \epsilon \sum_i \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) K_i + \frac{\partial L}{\partial \dot{q}_i} \frac{dK_i}{dt} \right] \\ &= \epsilon \frac{d}{dt} \left(\sum_i \frac{\partial L}{\partial \dot{q}_i} K_i \right). \end{aligned} \quad (1.4.11)$$

2.4 · Step 4 — recognise a total derivative and compare

The second line of (1.4.11) is the product rule read backwards. That is the whole trick: *the equation of motion is exactly what is needed to turn δL into a total time derivative of something*.

But the symmetry hypothesis says δL is *also* a total time derivative, namely $\epsilon \frac{dF}{dt}$. Two expressions for the same quantity:

$$\epsilon \frac{d}{dt} \left(\sum_i \frac{\partial L}{\partial \dot{q}_i} K_i \right) = \epsilon \frac{dF}{dt} \quad \implies \quad \frac{d}{dt} \left(\sum_i \frac{\partial L}{\partial \dot{q}_i} K_i - F \right) = 0. \quad (1.4.12)$$

Divide by ϵ and the theorem is proved:

$$Q = \sum_i p_i K_i - F, \quad \frac{dQ}{dt} = 0 \quad (1.4.13)$$

with $p_i = \partial L / \partial \dot{q}_i$ the canonical momentum of Chapter 1.2. Q is the **Noether charge**. ■

△ WHY THIS ISN'T OBVIOUS

Count the hypotheses and where each was spent, because the popular one-line version of this theorem — "symmetry implies conservation" — is missing one of them.

The symmetry hypothesis was used off shell. Step 2 computed δL for an arbitrary path. The definition in §1.2 demanded $\delta L = \epsilon \frac{dF}{dt}$ for *every* path, and that is what licenses comparing the two expressions in (1.4.12). A transformation that happens to leave L alone only on solutions tells you nothing; it is a statement about the solutions you already have, not an independent input.

The equations of motion were used, so the conclusion is on shell. Step 3 is where (1.4.10) entered, and it is indispensable — without it δL is not a total derivative of anything in particular. So Q is *not* constant along an arbitrary path. It is constant along paths that solve the equations of motion. "Energy is conserved" is a statement about trajectories the system actually takes, and it is silent about the wiggly competitor paths that Chapter 1.2 varied over. In the path integral of Chapter 5.8, where the competitor paths are physically present and contribute, this distinction stops being pedantic: conservation laws hold as statements about expectation values (Ehrenfest, Ward identities), not path by path.

Sharpen it into a slogan you can check: **the hypothesis is off shell, the conclusion is on shell**. Swap them and the theorem is either vacuous or false.

2.5 · Transformations that also move time

One case is not yet covered, and it happens to be the most important one. Time translation — "the laws are the same today as yesterday" — moves t itself, and the derivation above assumed t stood still. Most treatments wave at this step. It is worth doing, because the extra terms are exactly where the Hamiltonian comes from.

Allow the general case: a transformation that shifts the clock *and* the coordinates,

$$t \mapsto t' = t + \epsilon \tau(t), \quad q_i(t) \mapsto q'_i(t') = q_i(t) + \epsilon K_i. \quad (1.4.14)$$

Read the second equation carefully. It compares the new path at the *new* time with the old path at the *old* time — that is what a transformation of spacetime does. But Step 1 above needed a variation at fixed t . The grind box reconciles the two and grinds out the general result; here is the answer.

$$\boxed{Q = \sum_i p_i K_i - H \tau - F} \quad H \equiv \sum_i p_i \dot{q}_i - L. \quad (1.4.15)$$

The object H that appears attached to the time-shift τ is the combination Chapter 1.2 found by the Beltrami identity and Chapter 1.3 calls the **Hamiltonian**. It did not have to be put in by hand: it fell out of the bookkeeping of moving the clock. Setting $\tau = 0$ recovers (1.4.13), as it must.

▼ Grind box — the general variation, when the clock moves too

The two kinds of variation. Define the *total* variation, comparing new-at-new with old-at-old, and the *form* variation, comparing new-at- t with old-at- t :

$$\Delta q_i \equiv q'_i(t') - q_i(t) = \epsilon K_i, \quad \delta q_i \equiv q'_i(t) - q_i(t).$$

Only δ commutes with d/dt , so only δ can be fed into §2.1–§2.4. Relate them by Taylor expansion (Chapter 0.3), keeping first order in ϵ :

$$q'_i(t') = q'_i(t + \epsilon\tau) = q'_i(t) + \epsilon\tau \dot{q}'_i(t) + O(\epsilon^2) = q'_i(t) + \epsilon\tau \dot{q}_i(t) + O(\epsilon^2),$$

where in the last step $\dot{q}' = \dot{q} + O(\epsilon)$ was used, legitimate because it is already multiplied by ϵ . Rearranging,

$$\delta q_i = \epsilon(K_i - \dot{q}_i \tau) \equiv \epsilon \bar{K}_i.$$

The generator to use in the fixed- t derivation is not K but $\bar{K} = K - \dot{q} \tau$. That single substitution is the entire content of this grind box; the rest is showing what happens to the measure dt .

The action, with limits and measure transformed. The transformed action is an integral over the transformed interval:

$$S' = \int_{t'_1}^{t'_2} L\left(q'(s), \frac{dq'}{ds}(s), s\right) ds.$$

Substitute $s = t + \epsilon\tau(t)$, so that $ds = (1 + \epsilon\dot{\tau}) dt$ and the limits become t_1, t_2 again. Expanding the integrand about $s = t$ exactly as above — for any function G , $G(t + \epsilon\tau) = G(t) + \epsilon\tau \dot{G}(t) + O(\epsilon^2)$ — gives

$$\begin{aligned} S' &= \int_{t_1}^{t_2} \left[L(q'(t), \dot{q}'(t), t) + \epsilon\tau \frac{dL}{dt} \right] (1 + \epsilon\dot{\tau}) dt + O(\epsilon^2) \\ &= S + \int_{t_1}^{t_2} \left[\delta L + \epsilon\tau \frac{dL}{dt} + \epsilon\dot{\tau} L \right] dt + O(\epsilon^2) \\ &= S + \int_{t_1}^{t_2} \left[\delta L + \epsilon \frac{d}{dt}(\tau L) \right] dt + O(\epsilon^2), \end{aligned}$$

where δL is the fixed- t variation driven by $\bar{K}$, and the last line collected two terms into one by the product rule. So the natural statement of the symmetry hypothesis — *the action changes at most by a boundary term* — reads

$$\delta L + \epsilon \frac{d}{dt}(\tau L) = \epsilon \frac{dF}{dt}.$$

Finish. Steps 2 and 3 of §2 applied verbatim with generator $\bar{K}$ give, on shell, $\delta L = \epsilon \frac{d}{dt}(\sum_i p_i \bar{K}_i)$. Substituting,

$$\frac{d}{dt} \left(\sum_i p_i \bar{K}_i + \tau L - F \right) = 0.$$

Finally unpack $\bar{K}$ and recognise the Hamiltonian:

$$\sum_i p_i \bar{K}_i + \tau L = \sum_i p_i K_i - \tau \left(\sum_i p_i \dot{q}_i - L \right) = \sum_i p_i K_i - \tau H,$$

which is (1.4.15). ■

Why the sign of H looks backwards. In (1.4.15) the Hamiltonian enters with a minus sign, so pure time translation ($\tau = 1, K = 0, F = 0$) gives the conserved charge $Q = -H$. Since a constant multiple of a conserved quantity is conserved, that is the same statement as " H is conserved" — but the minus is not an accident, and §7 shows it is exactly the sign needed for H to generate *forward* time evolution.

3 · The three classics, each from one formula

Everything in this section is (1.4.15) with different entries. Nothing new is proved; the point is that three conservation laws with three unrelated textbook derivations are three substitutions into one equation.

3.1 · Time translation $\Rightarrow$ energy

Take $\tau = 1$ (shift every clock reading by the same amount) and $K_i = 0$ (do not touch the coordinates). By the grind box the form variation is $\bar{K}_i = -\dot{q}_i$, so

$$\delta L = -\epsilon \sum_i \left[\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right]. \quad (1.4.16)$$

Meanwhile the total time derivative of L along the path, by the chain rule, is

$$\frac{dL}{dt} = \sum_i \left[\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right] + \frac{\partial L}{\partial t}. \quad (1.4.17)$$

Add them as the symmetry condition of the grind box requires, $\delta L + \epsilon \frac{d}{dt}(\tau L) = \delta L + \epsilon \frac{dL}{dt}$, and the bracketed sums cancel exactly:

$$\delta L + \epsilon \frac{dL}{dt} = \epsilon \frac{\partial L}{\partial t}. \quad (1.4.18)$$

So time translation is a symmetry, with $F = 0$, **if and only if L has no explicit t -dependence**. Feeding $\tau = 1$, $K = 0$, $F = 0$ into (1.4.15):

$$Q = -H \quad \implies \quad E \equiv \sum_i p_i \dot{q}_i - L = \text{constant}. \quad (1.4.19)$$

That is the Beltrami identity of Chapter 1.2, Problem 2, which was obtained there by differentiating a lucky combination and watching terms cancel. It is also the Hamiltonian of Chapter 1.3, which was obtained there by a Legendre transform. Neither derivation said *why* that particular combination of p 's and $\dot{q}$'s should be the conserved one. This one does: it is the charge conjugate to moving the clock, and H appears multiplying τ in (1.4.15) for the same reason that p appears multiplying K .

For $L = \sum_a \frac{1}{2} m_a |\dot{\mathbf{x}}_a|^2 - V(\{\mathbf{x}\})$ the momentum is $p = m\dot{x}$ and $E = \sum_a m_a |\dot{\mathbf{x}}_a|^2 - (T - V) = T + V$: the familiar total energy. But note carefully what the theorem conserves. It conserves $\sum p\dot{q} - L$. Whether that equals $T + V$ is a separate question about the coordinates, and Chapter 1.2's Problem 3 (the bead on the motor-driven hoop) supplied a system where E is conserved and $T + V$ is not.

3.2 · Space translation $\Rightarrow$ momentum

Translate every particle by the same displacement, $\mathbf{x}_a \mapsto \mathbf{x}_a + \epsilon \mathbf{n}$, so $\mathbf{K}_a = \mathbf{n}$ for all a and $\tau = 0$. Since the velocities are untouched, only V can respond:

$$\delta L = -\epsilon \mathbf{n} \cdot \sum_a \frac{\partial V}{\partial \mathbf{x}_a}. \quad (1.4.20)$$

This vanishes for every $\mathbf{n}$ precisely when $\sum_a \partial V / \partial \mathbf{x}_a = 0$, which is the statement that V is unchanged by moving everything together — i.e. that V depends only on *differences* $\mathbf{x}_a - \mathbf{x}_b$. Then $F = 0$ and (1.4.15) gives

$$Q = \sum_a \mathbf{p}_a \cdot \mathbf{n} = \mathbf{P} \cdot \mathbf{n}, \quad \mathbf{P} = \sum_a m_a \dot{\mathbf{x}}_a. \quad (1.4.21)$$

Since $\mathbf{n}$ is arbitrary, all three components of $\mathbf{P}$ are conserved. Compare this with the Newtonian derivation, which invokes the third law and cancels internal forces in pairs. The third law is nowhere in sight here. What replaced it is the observation that a two-body potential can only depend on the separation — which is the same physical input, but now recognisable as a *symmetry* statement rather than an axiom about forces.

The one-coordinate version is worth naming because it is how you will use the theorem in practice: if some coordinate q_j is entirely absent from L (a **cyclic** or *ignorable* coordinate), then $\delta q_j = \epsilon$ is a symmetry with $F = 0$, and $Q = p_j$. That is the remark Chapter 1.2 made in passing after deriving $\dot{p}_i = \partial L / \partial q_i$; it is now a special case of a theorem.

3.3 · Rotation $\Rightarrow$ angular momentum

Rotate everything by ϵ about the fixed axis $\mathbf{n}$, so that from the table in §1.1

$$\delta \mathbf{x}_a = \epsilon \mathbf{n} \times \mathbf{x}_a, \quad \delta \dot{\mathbf{x}}_a = \epsilon \mathbf{n} \times \dot{\mathbf{x}}_a, \quad \tau = 0. \quad (1.4.22)$$

Check that this is a symmetry. The kinetic term responds as

$$\delta \left(\frac{1}{2} m \dot{\mathbf{x}} \cdot \dot{\mathbf{x}} \right) = \epsilon m \dot{\mathbf{x}} \cdot (\mathbf{n} \times \dot{\mathbf{x}}) = 0, \quad (1.4.23)$$

because a cross product is perpendicular to both its factors. The potential responds through the separations, and

$$\delta |\mathbf{x}_a - \mathbf{x}_b|^2 = 2 \epsilon (\mathbf{x}_a - \mathbf{x}_b) \cdot (\mathbf{n} \times (\mathbf{x}_a - \mathbf{x}_b)) = 0 \quad (1.4.24)$$

for the same reason. So if V depends only on the *lengths* of the separations — central forces — then $\delta L = 0$ exactly, with $F = 0$. Rotational symmetry is strict, not merely up to a total derivative. Now read off the charge, using the cyclic property of the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a})$:

$$Q = \sum_a \mathbf{p}_a \cdot (\mathbf{n} \times \mathbf{x}_a) = \mathbf{n} \cdot \sum_a (\mathbf{x}_a \times \mathbf{p}_a) = \mathbf{n} \cdot \mathbf{L}. \quad (1.4.25)$$

Arbitrary $\mathbf{n}$ again, so the whole vector $\mathbf{L} = \sum_a \mathbf{x}_a \times \mathbf{p}_a$ is conserved. And now the definition of angular momentum stops being an arbitrary-looking cross product that someone decided to write down: $\mathbf{x} \times \mathbf{p}$ is what you get when you contract the momentum with the generator of rotations. The cross product is there because the infinitesimal rotation was.

▼ Grind box — rotations in index notation, and where ε_{ijk} comes from

The vector notation above hides the structure that generalises. Write the rotation generator in components using the Levi-Civita symbol ε_{ijk} of Chapter 0.7 (antisymmetric in every pair, $\varepsilon_{123} = +1$):

$$\delta x_i = \epsilon \varepsilon_{ijk} n_j x_k, \quad \delta \dot{x}_i = \epsilon \varepsilon_{ijk} n_j \dot{x}_k,$$

with repeated indices summed. The kinetic term:

$$\delta \left(\frac{1}{2} m \dot{x}_i \dot{x}_i \right) = \epsilon m \dot{x}_i \varepsilon_{ijk} n_j \dot{x}_k = 0,$$

because ε_{ijk} is antisymmetric under $i \leftrightarrow k$ while $\dot{x}_i \dot{x}_k$ is symmetric, and the full contraction of a symmetric with an antisymmetric pair vanishes identically. That two-line argument is worth internalising: it is the reason rotations preserve lengths, and in Chapter 2.3 the same manoeuvre with a different symbol shows that Lorentz transformations preserve the spacetime interval.

The charge:

$$Q = p_i \varepsilon_{ijk} n_j x_k = n_j \varepsilon_{jki} x_k p_i = n_j L_j, \quad L_j \equiv \varepsilon_{jki} x_k p_i,$$

where the middle step used $\varepsilon_{ijk} = \varepsilon_{jki}$ (a cyclic permutation of three indices is even). Choosing $\mathbf{n} = \hat{\mathbf{z}}$ picks out

$$L_3 = \varepsilon_{312} x_1 p_2 + \varepsilon_{321} x_2 p_1 = x p_y - y p_x,$$

the familiar L_z .

The three rotations do not commute, and that matters. Rotating about x then y is not the same as y then x ; the discrepancy is, at second order, a rotation about z . Since the generators are what survive at first order, that failure to commute must show up as an algebraic relation among L_x, L_y, L_z — and §7 computes it: $\{L_i, L_j\} = \varepsilon_{ijk} L_k$. Translations, by contrast, commute with each other in any order, and correspondingly $\{P_i, P_j\} = 0$. The conserved charges inherit the *group structure* of the symmetries that produced them, which is the subject of Chapter 6.1 and the reason quantum angular momentum is quantised in Chapter 4.7.

3.4 · The table, and what it means

SYMMETRY OF THE ACTION	GENERATOR	CONSERVED CHARGE
Time translation, $t \rightarrow t + \epsilon$	$\tau = 1$	Energy $H = \sum p_i \dot{q}_i - L$
Space translation along $\mathbf{n}$	$\mathbf{K} = \mathbf{n}$	Momentum $\mathbf{n} \cdot \mathbf{P}$
Rotation about $\mathbf{n}$	$\mathbf{K} = \mathbf{n} \times \mathbf{x}$	Angular momentum $\mathbf{n} \cdot \mathbf{L}$
Galilean boost by $\mathbf{u}$	$\mathbf{K} = \mathbf{u} t, F = M \mathbf{u} \cdot \mathbf{R}_{\text{cm}}$	$\mathbf{P} t - M \mathbf{R}_{\text{cm}}$ (Worked Ex. 1)
Phase rotation $\phi \rightarrow e^{i\alpha} \phi$	$K = i\phi$	Electric charge (§6)

Read the first three rows as physics rather than as bookkeeping. The hypothesis in row one is that the laws are the same at every moment. The hypothesis in row two is that they are the same at every place. The hypothesis in row three is that they are the same in every direction. So:

WHAT THE CLASSICAL CONSERVATION LAWS ARE ACTUALLY ABOUT

Momentum is conserved because space is uniform. Energy is conserved because time is uniform. Angular momentum is conserved because space is isotropic.

None of these is a fact about matter. You cannot invent a new particle, or a new force, or a new material, that violates them — as long as it lives in a uniform, isotropic space and an unchanging time, its action has those symmetries and the charges are conserved whatever else it does. Conversely, the *only* way to break one is to break the corresponding symmetry, and §4 shows three situations where that genuinely happens. This is why conservation laws survived the transitions to relativity and quantum mechanics untouched while nearly everything else about mechanics was rewritten: they were never claims about mechanics in the first place.

4 · When symmetry fails, conservation fails

A theorem earns trust by being falsifiable. If symmetry really is the reason for conservation, then removing the symmetry must remove the conservation law — and remove *that one only*, leaving the others intact. This section takes the theorem at its word.

4.1 · The rate at which conservation fails

Start with the quantitative version, which is stronger than "the theorem no longer applies". Differentiate $E = \sum_i p_i \dot{q}_i - L$ along a solution, with L now allowed an explicit t -dependence:

$$\begin{aligned} \frac{dE}{dt} &= \sum_i \left[\dot{p}_i \dot{q}_i + p_i \ddot{q}_i \right] - \sum_i \left[\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right] - \frac{\partial L}{\partial t} \\ &= \sum_i \dot{q}_i \underbrace{\left[\dot{p}_i - \frac{\partial L}{\partial q_i} \right]}_{= 0 \text{ on shell}} + \sum_i \ddot{q}_i \underbrace{\left[p_i - \frac{\partial L}{\partial \dot{q}_i} \right]}_{= 0 \text{ by definition}} - \frac{\partial L}{\partial t}, \end{aligned} \quad (1.4.26)$$

$$\boxed{\frac{dE}{dt} = -\frac{\partial L}{\partial t}}. \quad (1.4.27)$$

Read that as an accounting identity. The left side is the rate at which the conservation law is violated; the right side is the rate at which the symmetry is violated. They are equal. Noether's theorem is the special case in which both sides are zero.

Example. A mass on a spring being driven by an external agent, $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2 + x f(t)$, has $\partial L / \partial t = x \dot{f}$, so $\frac{dE}{dt} = -x \dot{f}(t)$. The energy of the oscillator is not conserved — of course not, something outside the system is doing work on it. The Lagrangian's explicit t -dependence is the mathematical signature of "there is an outside". Enlarge the system to include whatever is driving f , and the enlarged L has no explicit t , and total energy is conserved again. Every apparent violation of energy conservation in ordinary physics is this: an incomplete system.

4.2 · A crystal: continuous symmetry broken to discrete

Now a case where the symmetry is not destroyed but *reduced*. An electron moving through a crystal lattice sees a potential that repeats:

$$V(\mathbf{x} + \mathbf{a}) = V(\mathbf{x}) \quad \text{for every lattice vector } \mathbf{a}, \quad (1.4.28)$$

but $V(\mathbf{x} + \delta) \neq V(\mathbf{x})$ for a general small displacement δ . Translation invariance survives only for a *discrete* set of displacements.

Noether's theorem has nothing to say. Its hypothesis was a family $q(t, \epsilon)$ continuous in ϵ , so that $\partial/\partial\epsilon$ exists at $\epsilon = 0$; a discrete symmetry has no ϵ to differentiate with respect to, and there is no generator K to put into (1.4.15). And indeed momentum is *not* conserved: the lattice exerts forces on the electron, and $\mathbf{p}$ changes.

▀ What survives is quoted, not derived here, and belongs to solid-state physics: because the symmetry group is still nontrivial, the eigenstates can be labelled by a quantity $\hbar\mathbf{k}$ called the **crystal momentum**, and it is conserved *modulo* a reciprocal lattice vector $\mathbf{G}$ — collisions can change $\mathbf{k}$ by any $\mathbf{G}$ at no cost. The pattern is general and worth carrying: a continuous symmetry gives an additively conserved quantity taking values in $\mathbb{R}$; breaking it to a discrete subgroup leaves a quantity conserved only up to the discrete "dual" of that subgroup. The residual freedom is exactly the amount of symmetry you threw away.

4.3 · An expanding universe: no conserved total energy

The striking case, and the one where the honest answer is more interesting than the tidy one.

▀ Quoting forward to Chapter 3.8: on the largest scales the universe is well described by a spacetime whose spatial distances all grow by a common factor $a(t)$, the scale factor. Matter fields living in that spacetime have an action whose Lagrangian contains $a(t)$ explicitly. So $\partial L/\partial t \neq 0$ — not because of some external driving agent that could be absorbed into a larger system, but because the geometry itself is time-dependent.

By (1.4.27) there is then no conserved energy, and this is not a technicality that a cleverer definition repairs. The most visible consequence is one you have heard described in a misleading way. Light from a distant galaxy arrives redshifted: its wavelength has been stretched by the factor a , so each photon's energy has fallen by $1/a$. The energy is not transferred to the expansion, and it is not stored anywhere. There is **no conserved global energy in this spacetime for it to be a part of**, so there is nothing that needs to balance. Asking where it went presupposes the very conservation law that the absence of time-translation symmetry has removed.

△ SAY EXACTLY THIS MUCH AND NO MORE

Three claims, in decreasing order of confidence.

What is certainly true. ▀ In a general curved spacetime there is no time-translation symmetry, hence no Noether charge for it, hence no meaningful "total energy of the universe". Chapter 3.5 makes this precise: the symmetries of a spacetime are its *Killing vector fields*, a conserved energy requires a timelike one, and the expanding solutions of Chapter 3.8 do not have one. Killing vectors turn out to be Noether's theorem written geometrically, which is why that chapter is where this loose end gets tied.

What survives. ▀ The *local* statement does hold, always and exactly: $\nabla_\mu T^{\mu\nu} = 0$, the covariant version of Chapter 0.7's continuity equation for the energy-momentum tensor. Energy and momentum are not created or destroyed at any point. But covariant conservation cannot be integrated over a large region to give a constant the way $\partial_\mu J^\mu = 0$ can — the extra connection terms of Chapter 3.3 obstruct exactly that step, and ν is a free index, so the object you would like to add up is a vector at each point and there is no coordinate-free way to add vectors at different points on a curved manifold (Chapter 3.2). Local conservation is real; global conservation is a boundary condition, and here the boundary conditions are not available.

What is true only sometimes. ▀ In spacetimes that are asymptotically flat — an isolated star, a black hole in an otherwise empty universe — the symmetry is restored far away, and a conserved total energy can be defined (the ADM mass). The expanding universe is not of this type. So "energy conservation fails in general relativity" is too strong, and "energy is always conserved, you just have to look harder" is false. The correct statement is the one Noether's theorem hands you directly: **you get a conserved energy exactly when you have a time-translation symmetry, and not otherwise.**

4.4 · Breaking one symmetry at a time

The figure below is the theorem's own falsification test. A particle orbits in a central potential; two dials sit next to it, and each breaks exactly one symmetry.

The system is a particle of mass m in the plane, with

$$V(r, \theta, t) = \frac{1}{2}m\omega^2 r^2 (1 + \lambda \cos 2\theta) (1 + \mu \sin \Omega t). \quad (1.4.29)$$

At $\lambda = \mu = 0$ this is the isotropic harmonic oscillator: rotationally symmetric, so L_z is conserved, and time-independent, so E is conserved. The dial λ squashes the potential into an ellipse — *rotational* symmetry gone, time-translation symmetry untouched. The dial μ makes the whole well breathe in time — *time-translation* symmetry gone, rotational symmetry untouched.

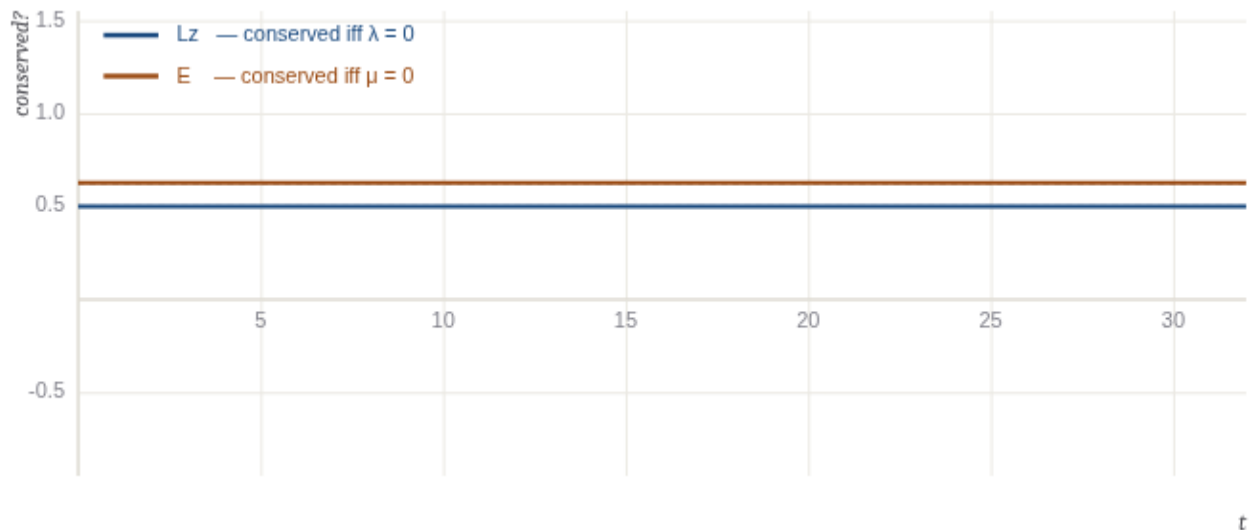
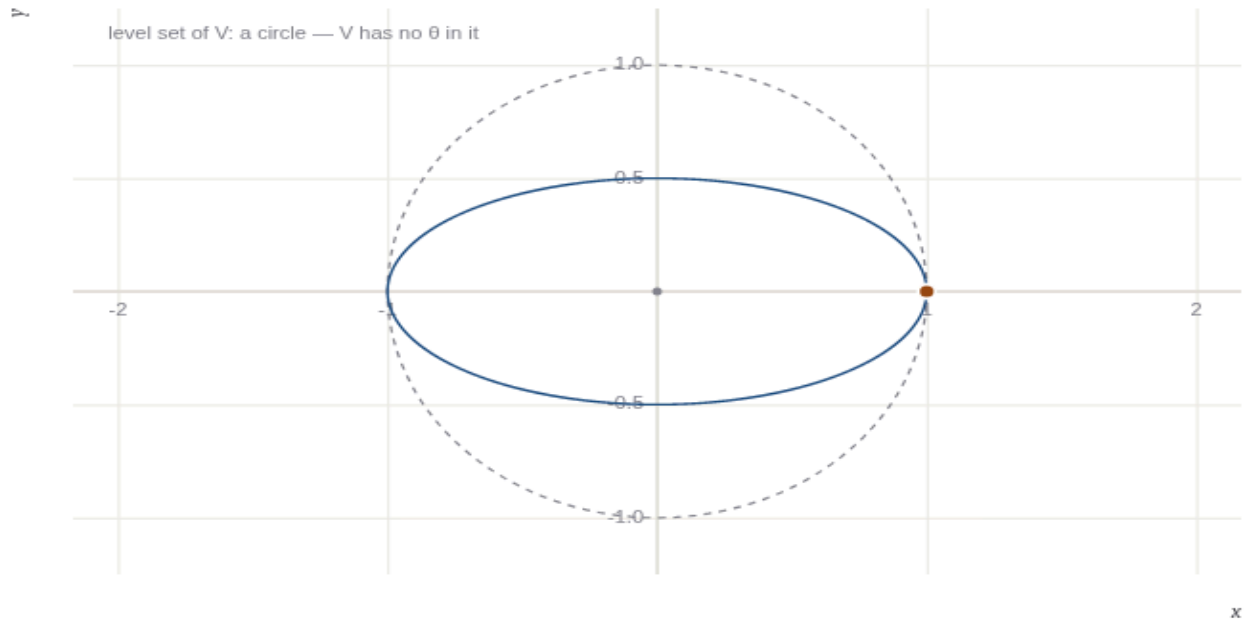
Before looking, extract a prediction. The angular equation of motion is $\frac{d}{dt}(mr^2\dot{\theta}) = -\partial V / \partial \theta$, and $mr^2\dot{\theta} = L_z$, so

$$\frac{dL_z}{dt} = -\frac{\partial V}{\partial \theta} = \lambda m \omega^2 r^2 \sin 2\theta (1 + \mu \sin \Omega t), \quad (1.4.30)$$

while (1.4.27) gives

$$\frac{dE}{dt} = \frac{\partial V}{\partial t} = \mu \Omega \cos \Omega t \cdot \frac{1}{2} m \omega^2 r^2 (1 + \lambda \cos 2\theta). \quad (1.4.31)$$

Each rate is *exactly proportional to its own dial* and completely independent of the other. The theorem does not merely predict that conservation fails; it predicts which one fails, and how fast.



λ breaks rotation $\lambda = 0.000$

μ breaks time-shift $\mu = 0.000$

both to zero

$\max |\Delta L_z| = 2.3e-13$

$\max |\Delta E| = 2.9e-13$

One dial, one conservation law. Units $m = \omega = 1$; the particle starts at $(x, y) = (1, 0)$ with velocity $(0, 0.5)$ and is integrated by fourth-order Runge–Kutta for 32 time units (about five orbits) at step 0.004. *Top*: the trajectory, with one level set of V dashed — normalised to enclose a fixed area, so that only its *shape* responds to the dial: a circle when $\lambda = 0$, an ellipse otherwise. That is the broken symmetry made visible. *Bottom*: $L_z(t)$ in blue and $E(t)$ in orange, against dashed lines at their initial values. With both dials at zero, both traces are flat to 3×10^{-13} — that residue is double-precision round-off in the integrator, not physics. Push λ to 0.005 — a half-percent squash — and L_z departs by 1.2% while E is still flat to 3×10^{-13} : ten orders of magnitude separate the broken law from the intact one. Push μ to 0.02 instead and the roles swap exactly, E wandering by 2% with L_z flat to 2×10^{-13} . Turn both up and both fail. There is no setting in which the wrong one fails.

One detail in the figure repays attention. At $\lambda = 0$ the orbit is a closed ellipse; the particle retraces its own path exactly. Turn λ up and the ellipse begins to precess and the path never closes. Closure of orbits is not a generic property — it is a symptom of extra conserved quantities, and Worked Example 2 is about the most famous case of that.

5 · Discrete symmetries — an honest exception

Nature has symmetries that are not continuous, and the theorem simply does not reach them. The three that matter are **parity** P (reflect all spatial coordinates, $\mathbf{x} \rightarrow -\mathbf{x}$), **charge conjugation** C (exchange every particle for its antiparticle), and **time reversal** T ($t \rightarrow -t$). None of them can be applied *a little bit*. Parity is the cleanest case to see why: the matrix $-I$ has determinant -1 in three dimensions, the identity has determinant $+1$, and the determinant of a rotation-or- reflection is a continuous function taking only those two values — so no continuous path of transformations connects P to the identity without leaving the group. There is therefore no one-parameter family to differentiate: no ϵ , no generator K , no (1.4.15), and no conserved current. What a discrete symmetry gives instead — and only in quantum mechanics, where states can be eigenvectors of the operation — is a *multiplicative* quantum number, typically ± 1 , whose product over the particles present is the same before and after a reaction. Additive charges come from continuous symmetries; multiplicative ones from discrete symmetries. They are different objects with different arithmetic.

■ Quoted, and developed in Chapter 6.6: both P and CP are *violated* in nature. Parity was found to fail in the weak interaction in 1957, and the combination CP was found to fail in 1964. This was a genuine surprise, and it is instructive about how much of physics is convention and how much is fact — nobody had ever suspected that the laws could distinguish left from right, because there is no reason within mechanics or electromagnetism why they should. There is no Noether charge to lose, and correspondingly no continuity equation quietly breaks; the symmetry just is not there. The one combination that appears to survive is CPT , and ■ that one is not an accident — it is a theorem, forced by Lorentz invariance and locality, which is why it has a different status from the others.

6 · Fields: symmetry gives a conserved *current*

Chapter 1.2 §8 showed that the Euler–Lagrange derivation goes through verbatim for fields, with one substitution: replace $L(q, \dot{q}, t)$ by a Lagrangian density $\mathcal{L}(\phi, \partial_\mu \phi)$, replace $\int dt$ by $\int d^4x$, and replace $\frac{d}{dt}$ by ∂_μ . The same substitution turns §2 into the field version of Noether's theorem, and the result is more useful than the mechanics version.

Take an **internal** symmetry — one that transforms the field at each point without moving the point:

$$\delta\phi = \epsilon K(\phi), \quad \delta(\partial_\mu \phi) = \partial_\mu(\delta\phi), \quad \delta\mathcal{L} = \epsilon \partial_\mu F^\mu. \quad (1.4.32)$$

The middle equation is Step 1 again — variations commute with derivatives — and the right-hand one is the symmetry hypothesis, with the total time derivative promoted to a four-divergence, since that is now what integrates to a boundary term. Steps 2 and 3 are unchanged. Expand by the chain rule, then use the field Euler–Lagrange equation $\partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi)) = \partial\mathcal{L}/\partial\phi$:

$$\begin{aligned}
\delta\mathcal{L} &= \frac{\partial\mathcal{L}}{\partial\phi} \delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \partial_\mu(\delta\phi) \\
&= \epsilon \left[\partial_\mu \left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \right) K + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \partial_\mu K \right] = \epsilon \partial_\mu \left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} K \right].
\end{aligned} \tag{1.4.33}$$

Compare with the hypothesis and the four-divergence of the difference vanishes:

$$\boxed{j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} K - F^\mu, \quad \partial_\mu j^\mu = 0.} \tag{1.4.34}$$

6.1 · This is the continuity equation, and that is a promotion

Split $j^\mu = (j^0, \mathbf{j})$ into its time and space parts. Then $\partial_\mu j^\mu = \partial_t j^0 + \nabla \cdot \mathbf{j}$, and (1.4.34) is

$$\frac{\partial j^0}{\partial t} + \nabla \cdot \mathbf{j} = 0, \tag{1.4.35}$$

which is precisely the continuity equation of Chapter 0.7, with j^0 the density of the conserved stuff and $\mathbf{j}$ its current. Chapter 0.7 promised that Noether's theorem would deliver one of these; here it is. Integrating over a region V and applying the divergence theorem, exactly as in that chapter,

$$Q = \int_V j^0 d^3x \quad \implies \quad \frac{dQ}{dt} = - \oint_{\partial V} \mathbf{j} \cdot d\mathbf{A}. \tag{1.4.36}$$

Now notice how much stronger this is than the mechanics version. In §2 we proved that a number $Q(t)$ does not change. Here we have proved something local: the charge in any region changes *only* by flux through that region's boundary. Charge cannot vanish in London and reappear in Sydney with the books still balancing — to get from one to the other it must cross every surface in between. **Field conservation laws forbid teleportation; mechanical ones do not.** That is not a stylistic upgrade. It is the difference between a conservation law compatible with special relativity and one that is not, since a global bookkeeping rule requires a notion of "at the same instant everywhere" that Chapter 2.2 will destroy.

▼ Grind box — the current is not unique, and the ambiguity is useful

Suppose j^μ is conserved and let $\Sigma^{\nu\mu}$ be any field antisymmetric in its two indices, $\Sigma^{\nu\mu} = -\Sigma^{\mu\nu}$. Define

$$\tilde{j}^\mu = j^\mu + \partial_\nu \Sigma^{\nu\mu}.$$

Then $\partial_\mu \tilde{j}^\mu = \partial_\mu j^\mu + \partial_\mu \partial_\nu \Sigma^{\nu\mu} = 0$, because $\partial_\mu \partial_\nu$ is symmetric in $\mu\nu$ (equality of mixed partials, Chapter 0.6) while $\Sigma^{\nu\mu}$ is antisymmetric, and a symmetric object fully contracted with an antisymmetric one vanishes — the identical argument used in §3.3's grind box. The new current is conserved too.

Does it carry the same charge? The added piece contributes $\int \partial_\nu \Sigma^{\nu 0} d^3x$. The $\nu = 0$ term vanishes by antisymmetry ($\Sigma^{00} = 0$), leaving $\int \partial_i \Sigma^{i0} d^3x$, a total spatial divergence, which by the divergence theorem is a surface integral at infinity. If the fields fall off, it is zero and $\tilde{Q} = Q$. So the current is ambiguous but the charge is not.

This is not idle. The Noether procedure applied to spacetime translations produces an energy-momentum tensor $T^{\mu\nu}$ that is in general *not symmetric* in its indices, which is a disaster for general relativity, where $T^{\mu\nu}$ sits on the right-hand side of a manifestly symmetric field equation (Chapter 3.6). The repair is exactly this freedom: one adds a suitable $\partial_\nu \Sigma^{\nu\mu\rho}$ — an "improvement term" — to symmetrise $T^{\mu\nu}$ without changing a single conserved charge. Chapter 5.2 does it.

6.2 · The example that sets up Part VI: a global phase

Take a *complex* scalar field ϕ , with ϕ and its conjugate ϕ^* treated as independent (which is legitimate for the same reason x and $\dot{x}$ were in Chapter 1.2 — they are independent slots of the function, and any complex ϕ can be traded for the real pair $(\text{Re } \phi, \text{Im } \phi)$ and back):

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi. \quad (1.4.37)$$

Now rotate the phase of the field by the same angle everywhere:

$$\phi \mapsto e^{i\alpha} \phi, \quad \phi^* \mapsto e^{-i\alpha} \phi^*, \quad \alpha \text{ a real constant.} \quad (1.4.38)$$

Every term in (1.4.37) pairs a ϕ with a ϕ^* , so the phases cancel exactly: $\mathcal{L}$ is strictly invariant, $F^\mu = 0$. This is a **global** $U(1)$ **symmetry** — $U(1)$ because the transformations are the unit complex numbers $e^{i\alpha}$, which form a circle group under multiplication, and *global* because α is the same at every point of spacetime.

Infinitesimally, $\epsilon = \alpha$ and $K_\phi = i\phi$, $K_{\phi^*} = -i\phi^*$. The two derivatives needed are $\partial \mathcal{L} / \partial(\partial_\mu \phi) = \partial^\mu \phi^*$ and $\partial \mathcal{L} / \partial(\partial_\mu \phi^*) = \partial^\mu \phi$, so (1.4.34) summed over both fields gives

$$j^\mu = i(\phi \partial^\mu \phi^* - \phi^* \partial^\mu \phi). \quad (1.4.39)$$

Two checks. It is real: complex-conjugating swaps the two terms and the i out front, giving back the same expression. And it is conserved directly, without invoking the theorem — take the divergence, note that the terms with two derivatives on different factors cancel in pairs, and use the field equation $\square \phi = -m^2 \phi$ from (1.4.37):

$$\partial_\mu j^\mu = i(\phi \square \phi^* - \phi^* \square \phi) = i(-m^2 \phi \phi^* + m^2 \phi^* \phi) = 0. \quad (1.4.40)$$

Multiplying a conserved current by any constant leaves it conserved, so the normalisation and overall sign are conventions; fixing them by the requirement that a positive-energy mode carry positive charge, the physical

electromagnetic current of a field of charge q is $-q j^\mu$. The conserved quantity $Q = \int j^0 d^3x$ is, up to that constant, **electric charge**.

THE QUESTION YOU SHOULD BE HOLDING

The symmetry above required α to be the *same everywhere*. That is a peculiar demand. It says that if you rotate the phase of the field in this room you must simultaneously rotate it, by exactly the same angle, in a galaxy ten billion light-years away — and Chapter 2.2 is about to make "simultaneously" a suspect word. A relativist's instinct is that a symmetry requiring instantaneous agreement across the universe is not a symmetry that should be taken seriously in that form.

So: what happens if you insist that α may depend on position, $\alpha \rightarrow \alpha(x)$?

The answer is that (1.4.37) is no longer invariant — the derivative ∂_μ hits $\alpha(x)$ and leaves a term behind — and the only way to repair it is to introduce a new field whose transformation law is designed to cancel that term. The new field turns out to be the electromagnetic potential A_μ , and its dynamics turn out to be Maxwell's equations. Demanding that a global symmetry be made local is called the **gauge principle**, and it does not merely accommodate electromagnetism — it *generates* it, and with a bigger group it generates the strong and weak forces as well. That is Chapter 6.3, and it is the single largest cheque this book writes. Nothing more will be said about it here; the point is only that you now know exactly which question leads there.

■ One warning to file, since it explains why Chapter 6.3 cannot simply run §6 again. Noether proved *two* theorems in 1918. The first is the one above, for symmetries with finitely many parameters. The second concerns symmetries depending on arbitrary *functions* — gauge symmetries — and its conclusion is different in kind: instead of conserved charges it yields identities among the equations of motion, which is why gauge theories have constraints, why the Bianchi identity of Chapter 3.4 exists, and why $\nabla_\mu T^{\mu\nu} = 0$ is automatic in general relativity rather than imposed.

7 · The other direction: charges generate their own symmetries

So far the arrow runs one way: symmetry $\Rightarrow$ conserved charge. This section runs it backwards, and the result is the real destination of Part I.

Chapter 1.3 supplies the tool. On phase space, coordinates q_i and momenta p_i are independent, and any two functions of them have a **Poisson bracket**

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right), \quad (1.4.41)$$

with the two facts we need: time evolution is $\frac{df}{dt} = \{f, H\} + \partial f / \partial t$, and a function G *generates* an infinitesimal transformation of phase space through

$$\delta f = \epsilon \{f, G\}. \quad (1.4.42)$$

7.1 · The charge is the generator — in one line

Take a Noether charge of the standard form (1.4.15) with $\tau = 0$, namely $Q = \sum_i p_i K_i(q, t) - F(q, t)$, and ask what transformation it generates. Only the first term contains any p , and it is linear in them, so

$$\delta q_j = \epsilon \{q_j, Q\} = \epsilon \frac{\partial Q}{\partial p_j} = \epsilon K_j. \quad (1.4.43)$$

That is not an analogy or a coincidence: it is an identity, and it says that **the Noether charge of a symmetry, used as a generator, reproduces exactly the symmetry it came from**. The theorem and its converse are the same equation read in opposite directions. Three instances, worth doing explicitly because the brackets are how you will compute in Part IV.

Momentum generates translations. With $G = p_x$, equation (1.4.41) gives $\{x, p_x\} = 1$ and $\{y, p_x\} = \{p_i, p_x\} = 0$, so

$$\delta x = \epsilon, \quad \delta y = \delta z = 0, \quad \delta \mathbf{p} = 0. \quad (1.4.44)$$

A rigid translation along x , exactly as advertised.

Angular momentum generates rotations. With $G = L_z = xp_y - yp_x$, the derivatives are $\partial L_z / \partial p_x = -y$, $\partial L_z / \partial p_y = x$, $\partial L_z / \partial x = p_y$, $\partial L_z / \partial y = -p_x$, so

$$\begin{aligned} \delta x &= \epsilon \{x, L_z\} = -\epsilon y, & \delta y &= \epsilon \{y, L_z\} = +\epsilon x, \\ \delta p_x &= \epsilon \{p_x, L_z\} = -\epsilon p_y, & \delta p_y &= \epsilon \{p_y, L_z\} = +\epsilon p_x. \end{aligned} \quad (1.4.45)$$

The position rotates by ϵ about the z -axis, and — a bonus the Lagrangian derivation never showed us — so does the momentum, which is exactly right, since momentum is a vector and vectors rotate.

The Hamiltonian generates time evolution. With $G = H$, equation (1.4.42) reads $\delta f = \epsilon \{f, H\} = \epsilon \frac{df}{dt}$ for any f without explicit time dependence, which is the statement that the transformation generated by H is *advance everything by ϵ* . Time evolution is not a different kind of process from a symmetry transformation; it is the symmetry transformation generated by the energy. (And this is where the minus sign in (1.4.15) pays: the Noether charge of time translation is $-H$, which generates $\delta f = -\epsilon \frac{df}{dt}$ — the shift of the graph of $q(t)$ forward by ϵ , which at fixed t shows you the value the old path had at $t - \epsilon$. The two signs describe the same motion from opposite sides.)

7.2 · The converse, and why it is not decorative

Now the reverse implication. Suppose Q is conserved and has no explicit time dependence. Then $0 = \frac{dQ}{dt} = \{Q, H\}$, and therefore the transformation Q generates does this to the Hamiltonian:

$$\delta H = \epsilon \{H, Q\} = -\epsilon \{Q, H\} = 0. \quad (1.4.46)$$

The Hamiltonian is invariant under the transformation generated by Q . So Q is not merely *produced by* a symmetry; Q is a symmetry, viewed as a generator. There is nothing left to prove in either direction.

THE DESTINATION OF PART I

Symmetries and conservation laws are not cause and effect. They are the same object, viewed twice. Given a symmetry, (1.4.15) hands you a function on phase space. Given that function, (1.4.42) hands you back the symmetry. The Noether charge and the generator are one thing wearing two names, and which name you use depends only on whether you are asking "what stays the same?" or "what moves?"

This has structural consequences that outrun mechanics. The charges close under the bracket: §3.3's grind box promised, and §7.3 verifies, that $\{L_i, L_j\} = \varepsilon_{ijk} L_k$ — the bracket of two conserved charges is again a conserved charge, with coefficients fixed by how the symmetries themselves compose. That structure is a **Lie algebra**, and it is the algebra of the symmetry *group*. Chapter 6.1 studies it systematically; Chapter 4.7 finds the identical relation with commutators in place of brackets, $[\hat{L}_i, \hat{L}_j] = i\hbar \varepsilon_{ijk} \hat{L}_k$, and derives the entire quantum theory of angular momentum — including half-integer spin, which has no classical counterpart — from nothing but that algebra.

This is what licenses the sentence this book keeps repeating. *Pick a symmetry, write the most general action invariant under it, quantize.* Step one is now a precise instruction: name a Lie group. Step two is Chapter 1.2. Step three replaces $\{, \}$ by $\frac{1}{i\hbar} [,]$ and is Chapter 5.3. Every remaining part of this book is that recipe with a different group in step one.

7.3 · The algebra of the rotation charges

▼ Grind box — computing $\{L_i, L_j\}$ by hand

Take $L_x = yp_z - zp_y$ and $L_y = zp_x - xp_z$ and grind (1.4.41) over the three coordinate pairs. Only terms in which the same variable is differentiated in both factors survive. The nonzero partials are

$$\begin{aligned} \frac{\partial L_x}{\partial y} &= p_z, & \frac{\partial L_x}{\partial z} &= -p_y, & \frac{\partial L_x}{\partial p_y} &= -z, & \frac{\partial L_x}{\partial p_z} &= y, \\ \frac{\partial L_y}{\partial z} &= p_x, & \frac{\partial L_y}{\partial x} &= -p_z, & \frac{\partial L_y}{\partial p_z} &= -x, & \frac{\partial L_y}{\partial p_x} &= z. \end{aligned}$$

Assemble the bracket term by term. From the x -pair: $\partial_x L_x \partial_{p_x} L_y - \partial_{p_x} L_x \partial_x L_y = 0 \cdot z - 0 \cdot (-p_z) = 0$. From the y -pair: $p_z \cdot 0 - (-z)(0) = 0$. From the z -pair: $(-p_y)(-x) - (y)(p_x) = xp_y - yp_x$. Hence

$$\{L_x, L_y\} = xp_y - yp_x = L_z,$$

and by relabelling $x \rightarrow y \rightarrow z \rightarrow x$, which permutes the three definitions cyclically,

$$\{L_i, L_j\} = \varepsilon_{ijk} L_k.$$

(Verified symbolically for all nine pairs.) Two remarks. First, the right-hand side is not zero, and that non-vanishing is the algebraic residue of the geometric fact that rotations about different axes do not commute — the bracket of generators measures the failure of the transformations to commute. Second, the same computation for translations gives $\{P_i, P_j\} = 0$, because translations do commute. The bracket is a faithful record of the group.

One consequence you can already read off, and which Chapter 4.7 turns into the whole theory of spin: since $\{L_x, L_y\} \neq 0$, the three components of angular momentum are not simultaneously "diagonalisable" in the sense that will matter after quantization. You will be able to know $|\mathbf{L}|$ and one component, and no more.

△ WHY THIS ISN'T OBVIOUS

A symmetry of the *equations of motion* is not the same thing as a symmetry of the *action*, and only the second kind gives a Noether charge. This is the second hypothesis that popular statements of the theorem drop.

Take the free particle, $L = \frac{1}{2}m\dot{x}^2$, whose equation of motion is $m\ddot{x} = 0$. Now scale the coordinate, $x \mapsto (1 + \epsilon)x$, leaving t alone. The equation of motion is completely unbothered: it is linear and homogeneous, so if $x(t)$ is a solution then so is any multiple of it. Every solution maps to a solution. Yet the action is not invariant —

$$L \mapsto \frac{1}{2}m(1 + \epsilon)^2 \dot{x}^2 = L + 2\epsilon L + O(\epsilon^2),$$

so $\delta L = 2\epsilon L$, which is *not* a total time derivative for a general path (if it were, L would contribute nothing to the equations of motion, and it plainly does). So this is not a symmetry in the sense of §1.2, and the theorem does not apply.

Test the conclusion rather than trusting it. If the theorem *did* apply, the generator is $K = x$ and the charge would be $Q = px = m\dot{x}x$. Differentiate along a solution: $\frac{dQ}{dt} = m\dot{x}^2 + m\dot{x}\ddot{x} = m\dot{x}^2 \neq 0$. Not conserved. The theorem was right to refuse.

What such a transformation gives instead is a map from solutions to *other* solutions — useful information, but of a different type. Problem 4 develops the most famous example: the scaling that maps Kepler orbits to other Kepler orbits, which is not a symmetry of the action, has no conserved charge, and is nevertheless the origin of Kepler's third law.

8 · Worked examples

WORKED EXAMPLE 1 — THE GALILEAN BOOST, AND A CONSERVATION LAW NOBODY NAMES

§1.3 established that the boost (1.4.5) is a symmetry of the N -body Lagrangian (1.4.4), but only in the loosened sense: it changes L by the total derivative of $F = M \mathbf{u} \cdot \mathbf{R}_{\text{cm}}$. Now collect the charge. The ingredients are

$$\mathbf{K}_a = \mathbf{u} t, \quad \tau = 0, \quad F = \mathbf{u} \cdot \sum_a m_a \mathbf{x}_a,$$

and (1.4.15) gives

$$Q = \sum_a \mathbf{p}_a \cdot (\mathbf{u} t) - \mathbf{u} \cdot \sum_a m_a \mathbf{x}_a = \mathbf{u} \cdot (\mathbf{P} t - M \mathbf{R}_{\text{cm}}).$$

Since $\mathbf{u}$ was an arbitrary boost direction, all three components are conserved:

$$\boxed{\mathbf{N} = \mathbf{P} t - M \mathbf{R}_{\text{cm}} = \text{constant.}}$$

Check it directly. $\frac{d\mathbf{N}}{dt} = \mathbf{P} + t \frac{d\mathbf{P}}{dt} - M \dot{\mathbf{R}}_{\text{cm}}$. The middle term vanishes because $\mathbf{P}$ is conserved (§3.2 — the same potential-depends-on-differences hypothesis), and $M \dot{\mathbf{R}}_{\text{cm}} = \sum_a m_a \dot{\mathbf{x}}_a = \mathbf{P}$ by definition. So $\frac{d\mathbf{N}}{dt} = \mathbf{P} - \mathbf{P} = 0$. ✓

What it says. Rearrange:

$$\mathbf{R}_{\text{cm}}(t) = \frac{\mathbf{P}}{M} t - \frac{\mathbf{N}}{M}.$$

The centre of mass moves in a straight line at constant velocity, forever, whatever the particles inside are doing to each other. That is a statement everybody knows and almost nobody files under "conservation law", because the conserved quantity has no name and no symbol in the standard curriculum. It has one here: it is the Noether charge of the boost, and it sits in the table of §3.4 alongside energy and momentum with exactly the same status.

Two things worth noticing. First, $\mathbf{N}$ depends *explicitly* on t . Nothing in the theorem forbade that — K was allowed to depend on t , and here $\mathbf{K} = \mathbf{u} t$ does. A conserved quantity need not be a function of the instantaneous state alone; it is a combination of the state and the clock that happens to be constant. Second, this is the charge that has no analogue after Chapter 2. The Galilean group's boosts get replaced by Lorentz boosts, whose Noether charges are the components of a relativistic object (Chapter 2.5) with a rather different physical reading. The centre-of-mass theorem is a specifically Newtonian statement, and it is a good illustration of the general rule that *changing the symmetry group changes the list of conserved quantities*, which is the whole reason the rest of this book is organised by symmetry group.

WORKED EXAMPLE 2 — THE LAPLACE–RUNGE–LENZ VECTOR, AND A HIDDEN SYMMETRY

Take the Kepler problem: one particle of mass m in the attractive potential

$$V(r) = -\frac{k}{r}, \quad \text{so that} \quad \dot{\mathbf{p}} = -\frac{k}{r^2} \hat{\mathbf{r}}, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}.$$

Rotational symmetry gives $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ conserved (§3.3), and time-translation symmetry gives E conserved (§3.1). That is four constants; a generic central force gives no more. But the $1/r$ potential has a fifth, discovered and rediscovered so often it carries three names:

$$\mathbf{A} = \mathbf{p} \times \mathbf{L} - mk\hat{\mathbf{r}}$$

It is conserved, and the proof is three lines of vector algebra done in the grind box below. The key move is that the awkward-looking $\hat{\mathbf{r}} \times (\mathbf{r} \times \dot{\mathbf{r}})$ collapses, by the $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ expansion, into exactly $r^2 \frac{d}{dt}(\mathbf{r}/r)$ — so the rate of change of $\mathbf{p} \times \mathbf{L}$ is $mk \frac{d\hat{\mathbf{r}}}{dt}$, and subtracting $mk\hat{\mathbf{r}}$ cancels it.

▼ Grind box — proving $\frac{d\mathbf{A}}{dt} = 0$, and two identities that come free

Conservation. $\mathbf{L}$ is constant (§3.3), so only the first factor moves:

$$\frac{d}{dt}(\mathbf{p} \times \mathbf{L}) = \dot{\mathbf{p}} \times \mathbf{L} = -\frac{k}{r^2} \hat{\mathbf{r}} \times (m\mathbf{r} \times \dot{\mathbf{r}}).$$

Apply $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$ with $\mathbf{a} = \hat{\mathbf{r}}$, $\mathbf{b} = \mathbf{r}$, $\mathbf{c} = \dot{\mathbf{r}}$, and use $\hat{\mathbf{r}} \cdot \dot{\mathbf{r}} = \dot{r}$ together with $\hat{\mathbf{r}} \cdot \mathbf{r} = r$:

$$\begin{aligned} \dot{\mathbf{p}} \times \mathbf{L} &= -\frac{mk}{r^2} [\mathbf{r} \dot{r} - \dot{\mathbf{r}} r] = mk \left[\frac{\dot{\mathbf{r}}}{r} - \frac{\mathbf{r} \dot{r}}{r^2} \right] \\ &= mk \frac{d}{dt} \left(\frac{\mathbf{r}}{r} \right) = mk \frac{d\hat{\mathbf{r}}}{dt}, \end{aligned}$$

the last line by the quotient rule. Hence $\frac{d}{dt}(\mathbf{p} \times \mathbf{L} - mk\hat{\mathbf{r}}) = 0$. ■ (Verified symbolically in three dimensions, componentwise, on the equations of motion.)

Identity one: $\mathbf{A}$ lies in the orbital plane. Dot with $\mathbf{L}$. The first term gives $(\mathbf{p} \times \mathbf{L}) \cdot \mathbf{L} = 0$ because a cross product is perpendicular to its factors; the second gives $-mk\hat{\mathbf{r}} \cdot \mathbf{L} = 0$ because $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is perpendicular to $\mathbf{r}$. So $\mathbf{A} \cdot \mathbf{L} = 0$: the vector lies in the plane of the orbit, which is why it can point at the perihelion.

Identity two: its length is fixed by E and L . Square it, using $\mathbf{p} \perp \mathbf{L}$ (so $|\mathbf{p} \times \mathbf{L}| = pL$) and $(\mathbf{p} \times \mathbf{L}) \cdot \hat{\mathbf{r}} = \mathbf{L} \cdot (\mathbf{r} \times \mathbf{p})/r = L^2/r$:

$$\begin{aligned} A^2 &= p^2 L^2 + m^2 k^2 - 2mk \frac{L^2}{r} = L^2 \left(p^2 - \frac{2mk}{r} \right) + m^2 k^2 \\ &= 2mE L^2 + m^2 k^2, \end{aligned}$$

since $E = p^2/2m - k/r$ gives $p^2 - 2mk/r = 2mE$. So $\mathbf{A}$ is not a sixth independent constant: its direction in the plane is one new number and its length is determined. Five independent constants, not six — which is exactly the count needed to fix an orbit in space completely apart from where along it the particle currently sits.

What it is. Dot $\mathbf{A}$ into $\mathbf{r}$, using the cyclic property of the scalar triple product:

$$\mathbf{A} \cdot \mathbf{r} = (\mathbf{p} \times \mathbf{L}) \cdot \mathbf{r} - mkr = \mathbf{L} \cdot (\mathbf{r} \times \mathbf{p}) - mkr = L^2 - mkr.$$

Writing $\mathbf{A} \cdot \mathbf{r} = Ar \cos \theta$ with θ measured from the fixed direction of $\mathbf{A}$ and solving for r :

$$r(\theta) = \frac{L^2/mk}{1 + e \cos \theta}, \quad e \equiv \frac{A}{mk}.$$

That is the polar equation of a conic of eccentricity e — an ellipse for $e < 1$ — with r smallest at $\theta = 0$. So $\mathbf{A}$ **points from the focus to the perihelion, and its length is the eccentricity times mk** . Its conservation is the statement that the orbit's axis never moves: *Kepler ellipses close and do not precess*. Adding any deviation from exact $1/r$ destroys $\mathbf{A}$ and the orbit precesses — which is exactly what the λ dial did to the closed orbit in §4.4's figure, and exactly what the Sun's other planets, and then general relativity, do to Mercury (Chapter 3.7).

Where is the symmetry? By §7.2 there must be one: $\mathbf{A}$ has no explicit time dependence and is conserved, so $\{\mathbf{A}, H\} = 0$, so the transformation $\mathbf{A}$ generates leaves H invariant. But look at the list of §3.4 — nothing there produces $\mathbf{A}$. The reason is that the transformation generated by $\mathbf{A}$ is not a motion of space at all. It mixes positions with momenta, deforming an orbit into a different orbit of the same energy — turning a circle into a thin ellipse, say. No rearrangement of the *coordinates* does that, which is why it is called a **hidden** symmetry. The Lagrangian derivation of §2 assumed the transformation acted on q alone and could never have found it; the phase-space picture of §7 finds it immediately, because on phase space q and p are on the same footing and there are transformations that mix them.

Now the structure. The brackets of the six conserved quantities close on themselves — this part is a computation, of exactly the kind §7.3 did by hand, only longer:

$$\{L_i, L_j\} = \varepsilon_{ijk} L_k, \quad \{L_i, A_j\} = \varepsilon_{ijk} A_k, \quad \{A_i, A_j\} = -2mH \varepsilon_{ijk} L_k,$$

(all three verified symbolically here.) ■ The identification that follows is quoted, since naming an abstract Lie algebra is Chapter 6.1's business: for bound orbits $E < 0$ the rescaled vector $\mathbf{D} = \mathbf{A}/\sqrt{-2mE}$ makes the last relation match the first. The resulting six-generator Lie algebra is that of $SO(4)$ — the rotation group of *four*-dimensional space. The Kepler problem in three dimensions has the symmetry of rotations in four. That is the hidden symmetry, and it is why there are five constants instead of four and why the orbits close.

■ And the payoff arrives in Chapter 4.5. The hydrogen atom is the quantum $1/r$ problem, so it inherits this same $SO(4)$. Its energy levels depend only on the principal quantum number n and not on the orbital angular momentum ℓ — the $2s$ and $2p$ states are degenerate — which looks like a numerical accident and is presented as one in most first courses. It is not an accident. It is the extra symmetry, showing up as a degeneracy, exactly as the $\lambda = 0$ orbit in §4.4's figure showed its extra symmetry by closing.

Which is the moral, and it is worth carrying out of this chapter as a working heuristic: **an unexplained coincidence in physics is usually an unrecognised symmetry**. A degeneracy that "just happens", a quantity that "happens" to stay constant, a term that "happens" to cancel — each is a symmetry that has not yet been named. Looking for it is one of the most reliably productive things a physicist can do, and Chapter 6.6 is what happens when the search succeeds and Chapter 5.9 is what happens when it fails.

9 • Your turn

PROBLEM 1 — A SCALING THAT IS A NOETHER SYMMETRY

Consider a particle on the half-line $x > 0$ with

$$L = \frac{1}{2}m\dot{x}^2 - \frac{k}{x^2}, \quad k > 0.$$

(a) Show that the combined scaling $x \mapsto \ell x, t \mapsto \ell^2 t$ leaves the *action* exactly invariant. (b) Identify K and τ and use (1.4.15) to find the conserved charge. (c) Verify directly that it is conserved. (d) Why does this scaling work when §7's free-particle scaling did not?

– Solution

(a) Under $x'(t') = \ell x(t)$ with $t' = \ell^2 t$, the velocity scales as $dx'/dt' = \ell/\ell^2 \cdot \dot{x} = \ell^{-1} \dot{x}$, so the kinetic term scales by ℓ^{-2} ; the potential k/x'^2 scales by ℓ^{-2} as well. Both terms of L therefore carry the same factor ℓ^{-2} , while the measure $dt' = \ell^2 dt$ carries ℓ^{+2} :

$$S' = \int L' dt' = \int \ell^{-2} L \cdot \ell^2 dt = S.$$

Exact invariance, $F = 0$. (Confirmed symbolically.) The reason it works is that the two terms of L have the *same* scaling weight, and $1/x^2$ is the unique power for which they do. This is the simplest system in physics with a scale symmetry.

(b) Set $\ell = 1 + \epsilon$. Then $\delta x = \epsilon x$ and $\delta t = 2\epsilon t$, so $K = x$ and $\tau = 2t$, and (1.4.15) gives

$$Q = px - 2tH = m\dot{x}x - 2t\left(\frac{1}{2}m\dot{x}^2 + \frac{k}{x^2}\right).$$

(c) The equation of motion is $m\ddot{x} = -\frac{d}{dx}\left(\frac{k}{x^2}\right) = \frac{2k}{x^3}$. Differentiate Q , remembering that $H = E$ is itself conserved because L has no explicit t :

$$\frac{dQ}{dt} = m\dot{x}^2 + m\dot{x}\ddot{x} - 2E = m\dot{x}^2 + x \cdot \frac{2k}{x^3} - 2\left(\frac{1}{2}m\dot{x}^2 + \frac{k}{x^2}\right) = 0. \checkmark$$

Note the structure: the charge is explicitly time-dependent, like the boost charge of Worked Example 1, and for the same reason — τ and K contained t .

(d) Because here the *time* was scaled too, and by exactly the amount needed to make the measure compensate the integrand. In §7's free particle only x was scaled, giving $S \mapsto \ell^2 S$ — the equations of motion survive (they are unchanged by an overall constant factor on S) but the action does not, and the theorem's hypothesis fails. The lesson is that whether a scaling is a Noether symmetry depends on *how* it

acts on time, and finding the right relative weight is exactly what one does when constructing a scale-invariant theory.

▀ Forward pointer: this Lagrangian is the one-dimensional prototype of a **conformally invariant** theory. The full symmetry is larger than the single scaling — there are three charges, closing on the algebra of $SL(2, \mathbb{R})$ — and the two-dimensional version of the same structure, with infinitely many charges instead of three, is what makes string theory calculable in Chapter 7.2.

PROBLEM 2 — A PARTICLE ON A SURFACE OF REVOLUTION

A particle of mass m slides without friction on the surface obtained by rotating the curve $\rho = R(z)$ about the z -axis, with no potential ($V = 0$). Using cylindrical coordinates (ρ, φ, z) :

(a) Write L on the surface. (b) Identify the symmetry and its Noether charge. (c) Use it to reduce the motion to a single first-order equation for $z(t)$, i.e. to a quadrature. (d) Show that the charge is equivalent to *Clairaut's relation*: $R \sin \psi$ is constant, where ψ is the angle between the trajectory and the local meridian.

– Solution

(a) In cylindrical coordinates the squared speed is $\dot{\rho}^2 + \rho^2 \dot{\varphi}^2 + \dot{z}^2$. On the surface $\rho = R(z)$, so $\dot{\rho} = R'(z)\dot{z}$ and

$$L = \frac{1}{2}m \left[(1 + R'(z)^2) \dot{z}^2 + R(z)^2 \dot{\varphi}^2 \right].$$

Two coordinates, because imposing the constraint by choosing coordinates is exactly what Chapter 1.2 §6 recommended.

(b) φ does not appear in L — the surface is unchanged by rotating about its axis, which is the symmetry $\delta\varphi = \epsilon$, $K_\varphi = 1$, $\tau = 0$, $F = 0$. So

$$Q = \frac{\partial L}{\partial \dot{\varphi}} = m R(z)^2 \dot{\varphi} \equiv \ell = \text{constant},$$

the angular momentum about the axis. (This is §3.2's cyclic-coordinate case, and it is what Chapter 1.2 saw in polar coordinates without a name for it.)

(c) L has no explicit t , so $E = \frac{1}{2}m[(1 + R'^2)\dot{z}^2 + R^2\dot{\varphi}^2]$ is conserved as well. Eliminate $\dot{\varphi} = \ell/(mR^2)$:

$$E = \frac{1}{2}m(1 + R'^2)\dot{z}^2 + \underbrace{\frac{\ell^2}{2mR(z)^2}}_{V_{\text{eff}}(z)}.$$

Two conserved quantities have reduced two coupled second-order equations to one first-order equation, which separates:

$$t - t_0 = \int \sqrt{\frac{m(1 + R'(z)^2)}{2(E - \ell^2/2mR(z)^2)}} dz,$$

and then $\varphi(t)$ follows by integrating $\dot{\varphi} = \ell/mR^2$. The problem is solved by quadrature. *Every* conserved quantity buys one integration; two of them turn a two-degree-of-freedom problem into arithmetic.

Note also that $V_{\text{eff}} \propto 1/R^2$ acts as a barrier wherever the surface narrows: a particle with $\ell \neq 0$ cannot reach a neck of radius smaller than $R_{\text{min}} = \ell/\sqrt{2mE}$. That is the centrifugal barrier, arriving here as a consequence of a symmetry rather than as a fictitious force.

(d) With $V = 0$ the speed v is constant (energy conservation). Decompose the velocity into a component along the local circle of latitude, $v_\varphi = R\dot{\varphi}$, and one along the meridian. If ψ is the angle between the trajectory and the meridian, then $\sin \psi = v_\varphi/v$, so

$$\ell = mR^2\dot{\varphi} = mRv_\varphi = mvR\sin\psi \quad \implies \quad R\sin\psi = \text{constant}.$$

This is **Clairaut's relation**, and it is what Chapter 1.2's grind box on spherical geodesics found by integrating the Euler–Lagrange equation and then remarked was "Noether's theorem arriving three chapters early disguised as cartography". Here is the general version: on any surface of revolution, the geodesics obey $R\sin\psi = \text{constant}$ because the surface has an axial symmetry. It is also the reason a great-circle route from London to Tokyo goes over the Arctic — the path must turn toward the axis as R decreases to keep the product fixed.

PROBLEM 3 — THE CURRENT OF A COMPLEX FIELD, AND WHAT THE CHARGE IS

For the field theory of (1.4.37), derive the Noether current for the global phase symmetry from scratch (do not quote §6.2), and evaluate the charge density j^0 on the plane-wave solution $\phi = Ne^{-i(Et-\mathbf{p}\cdot\mathbf{x})}$ with N real. Then evaluate it on the complex-conjugate solution ϕ^* and say what the sign means.

– Solution

Setup. Treat ϕ and ϕ^* as two independent fields. Writing $\mathcal{L} = g^{\mu\nu}\partial_\mu\phi^*\partial_\nu\phi - m^2\phi^*\phi$,

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi^*, \quad \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)} = \partial^\mu\phi.$$

The transformation. $\phi \rightarrow e^{i\alpha}\phi$ gives, to first order in $\epsilon = \alpha$,

$$\delta\phi = i\alpha\phi \Rightarrow K_\phi = i\phi, \quad \delta\phi^* = -i\alpha\phi^* \Rightarrow K_{\phi^*} = -i\phi^*.$$

$\mathcal{L}$ pairs each ϕ with a ϕ^* , so the phases cancel and $\delta\mathcal{L} = 0$ exactly: $F^\mu = 0$.

The current. Sum (1.4.34) over the two fields:

$$j^\mu = \partial^\mu\phi^*(i\phi) + \partial^\mu\phi(-i\phi^*) = i(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi).$$

On a plane wave. With $\phi = Ne^{-i(Et-\mathbf{p}\cdot\mathbf{x})}$ we have $\partial^0\phi = \partial_t\phi = -iE\phi$ and $\partial^0\phi^* = +iE\phi^*$, so

$$j^0 = i(\phi(iE\phi^*) - \phi^*(-iE\phi)) = i(2iE|\phi|^2) = -2EN^2.$$

The conjugate solution. $\phi^* = Ne^{+i(Et-\mathbf{p}\cdot\mathbf{x})}$ is also a solution of $\square\phi = -m^2\phi$, and running the same computation gives $j^0 = +2EN^2$: the same magnitude, opposite sign.

What it means. A complex field carries a conserved charge, and the field has two families of solutions carrying that charge with opposite signs and equal magnitude. A *real* field has no such symmetry — there is no phase to rotate — and therefore no conserved charge at all. So: **a field carries charge if and only if it is complex.** Multiplying j^μ by the constant $-q$ (allowed, since any constant multiple of a conserved current is conserved) makes the first solution carry charge $+q$ per unit $2EN^2$ and the second $-q$.

► Forward: after quantization in Chapter 5.3 the two families become particles and *antiparticles*.

Antimatter is not an extra postulate — it is the second sign of the Noether charge of a $U(1)$ symmetry, and it exists for exactly the same reason the charge does. A neutral particle like the photon is described by a real field precisely because it has no $U(1)$ charge to carry, and is therefore its own antiparticle.

PROBLEM 4 — A SYMMETRY WITH NO CHARGE, AND KEPLER'S THIRD LAW

For the Kepler Lagrangian $L = \frac{1}{2}m|\dot{\mathbf{r}}|^2 + k/r$, consider the scaling

$$\mathbf{r} \mapsto \ell \mathbf{r}, \quad t \mapsto \ell^{3/2} t.$$

(a) Show that it maps solutions to solutions. (b) Show that it is *not* a symmetry of the action, and that the would-be Noether charge is not conserved. (c) What does the transformation give you instead? (d) Contrast with Problem 1.

– Solution

(a) The equation of motion is $m\ddot{\mathbf{r}} = -k\mathbf{r}/r^3$. Under the scaling, the left side picks up $\ell/\ell^3 = \ell^{-2}$ and the right side picks up $\ell/\ell^3 = \ell^{-2}$ as well. Both sides scale identically, so the equation is unchanged and every solution maps to a solution.

(b) Now scale the action. The velocity scales by $\ell/\ell^{3/2} = \ell^{-1/2}$, so $T \mapsto \ell^{-1}T$; the potential $k/r \mapsto \ell^{-1}k/r$. Both terms of L carry ℓ^{-1} , while dt carries $\ell^{3/2}$, so

$$S \mapsto \ell^{-1} \cdot \ell^{3/2} S = \ell^{1/2} S \neq S.$$

Not invariant, and not invariant up to a boundary term either — the change is proportional to S itself, which is not the integral of a total derivative for a general path. The hypothesis of §1.2 fails.

Test the conclusion. With $\ell = 1 + \epsilon$ the would-be generators are $\mathbf{K} = \mathbf{r}$ and $\tau = \frac{3}{2}t$, so (1.4.15) would offer

$$Q \stackrel{?}{=} \mathbf{p} \cdot \mathbf{r} - \frac{3}{2}tE.$$

Differentiate along a solution, using $\frac{d}{dt}(\mathbf{p} \cdot \mathbf{r}) = |\mathbf{p}|^2/m + \mathbf{r} \cdot \dot{\mathbf{p}} = 2T - k/r$ (the last step because $\mathbf{r} \cdot \dot{\mathbf{p}} = -k/r$):

$$\frac{dQ}{dt} = 2T - \frac{k}{r} - \frac{3}{2}E = 2T - \frac{k}{r} - \frac{3}{2}\left(T - \frac{k}{r}\right) = \frac{1}{2}\left(T + \frac{k}{r}\right) = \frac{1}{2}L \neq 0.$$

Not conserved — and not even conserved on average: over one closed orbit $\mathbf{p} \cdot \mathbf{r}$ returns to its starting value, so $\left\langle \frac{dQ}{dt} \right\rangle = -\frac{3}{2}E \neq 0$ for a bound orbit. What *does* average to zero is the simpler combination $G = \mathbf{p} \cdot \mathbf{r}$ on its own, whose rate is $\frac{dG}{dt} = 2T - k/r$; setting its average to zero gives

$$\langle 2T \rangle = \langle k/r \rangle,$$

the **virial theorem**. That is the residue a non-symmetry leaves behind: not a constant, but a relation between time-averages.

(c) The transformation relates *different* solutions rather than constraining a single one. Take any orbit, scale all lengths by ℓ and all times by $\ell^{3/2}$, and you get another legitimate orbit. Since the semi-major axis scales as $a \mapsto \ell a$ and the period as $T \mapsto \ell^{3/2}T$, the combination

$$\frac{T^2}{a^3} \mapsto \frac{\ell^3 T^2}{\ell^3 a^3} = \frac{T^2}{a^3}$$

is the same for every orbit. That is **Kepler's third law**, obtained without solving the equations of motion at all — from a symmetry that has no Noether charge. Symmetries of the equations of motion are not useless; they simply produce a different kind of information: relations *between* solutions, not constants *along* one.

(d) The two problems differ in exactly one respect: whether the time-scaling exponent is the one that makes $\int L dt$ balance. In Problem 1, L scaled by ℓ^{-2} and dt by ℓ^{+2} — they cancel, giving a Noether charge. Here L scales by ℓ^{-1} and dt by $\ell^{3/2}$ — they do not, and there is none. Same kind of transformation, same kind of system, and the presence or absence of a conservation law turns on a single exponent. That is a useful reminder that "is this a symmetry?" is a question about the action, and can only be answered by computing.

THE BRICK YOU JUST LAID

You can now convert symmetry into conservation, in either direction, by formula. Given a transformation $\delta q_i = \epsilon K_i$, $\delta t = \epsilon \tau$ under which the action is invariant up to a boundary term ϵF , the quantity $Q = \sum_i p_i K_i - H\tau - F$ is constant along solutions (1.4.15); and given a conserved Q , the bracket $\delta f = \epsilon \{f, Q\}$ hands the symmetry back. The proof was four steps — variations commute with d/dt , expand by the chain rule, substitute the Euler–Lagrange equation, recognise a total derivative — and the only subtlety worth remembering is that the hypothesis is off shell while the conclusion is on shell.

From the one formula came energy, momentum, angular momentum, the uniform motion of the centre of mass, and electric charge. You saw conservation fail, on screen, in exactly the component whose symmetry you broke, at exactly the predicted rate. You saw that discrete symmetries give no current, that in fields the theorem upgrades to a *local* statement — Chapter 0.7's continuity equation — and that an unexplained degeneracy is usually a symmetry nobody has named yet.

Where this gets spent. Chapter 2.5 applies it to spacetime translations in Minkowski space and gets four-momentum conservation as a single relation. Chapter 3.5 geometrises it: Killing vectors are Noether generators written as vector fields on a manifold, and Chapter 3.6 finds $\nabla_\mu T^{\mu\nu} = 0$ falling out of the invariance of the Einstein–Hilbert action under coordinate changes. Chapter 5.2 runs \$6 in earnest for relativistic fields and constructs the energy–momentum tensor. Chapter 6.3 takes \$6.2's $U(1)$, makes α depend on position, and out comes electromagnetism; Chapter 6.7 does it with the full Standard Model group. And Chapter 4.7 finds \$7's bracket algebra again with commutators, which is where spin comes from.

Part I, closed. Four chapters ago mechanics was a list of forces, one per interaction, each resolved into components in a coordinate system chosen for convenience. It is now four objects: a single scalar L ; the condition $\delta S = 0$ that turns it into equations of motion; a bracket $\{, \}$ that turns phase-space functions into motions; and a theorem converting symmetry into conservation. Nothing in that list mentions gravity, or electrons, or fields, or any particular physics at all — which is the point, because **every remaining part of this book is that same package applied to a different symmetry group.** Lorentz transformations in Part II. Diffeomorphisms — arbitrary smooth coordinate changes — in Part III. Unitary phase rotations, first global and then local, in Parts IV to VI. Conformal transformations of a two-dimensional worldsheet in Part VII. You now have the whole method. What remains is to learn the groups.